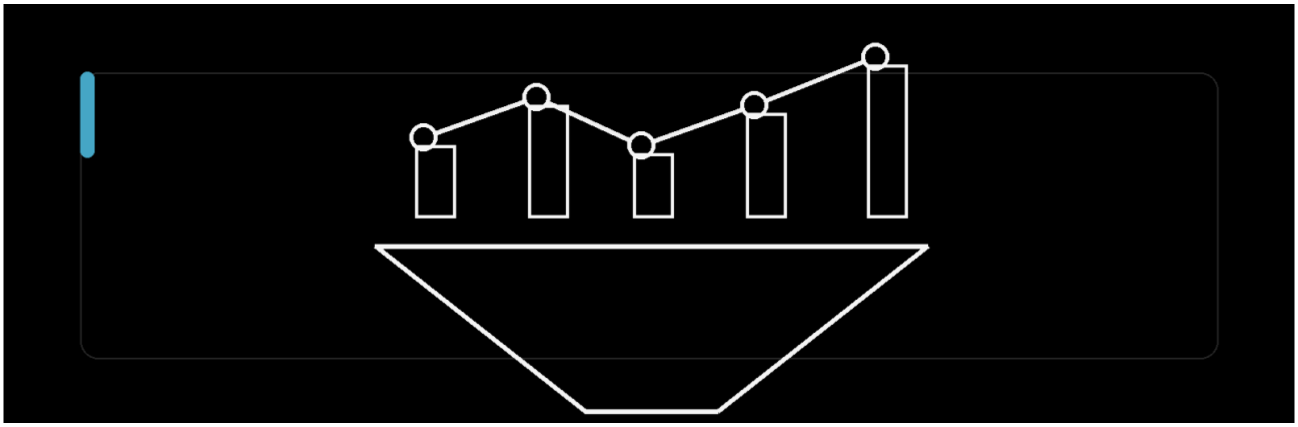

Funnel Analysis

Tags	EDA · Excel · Python · Funnel Analysis · Statistical Testing
Description	**Customer Journey Optimisation and Resource Prioritisation**

Executive Summary

I analysed **12,719 customer-journey events** across **5,000 sessions**. Of these, **1,010 sessions** ended in a purchase, giving an overall session purchase rate of **20.2%**.

The clearest bottleneck appeared during product evaluation. Of the 3,987 sessions that reached a product page, only 1,599 progressed to the cart. This represents a **59.9% drop-off** and a loss of **2,388 sessions** at one transition, making product page-to-cart progression the strongest improvement opportunity.

I also compared purchase outcomes across device, region, referral source and time of day. The Pearson chi-square tests did not identify statistically significant differences between these groups, so I prioritised improvements to the shared product experience rather than a broad segment-specific redesign.

Problem

The site attracts traffic, but most sessions do not progress to purchase. I wanted to identify where customers disengage, determine which patterns are supported by the event data, and decide where product and analytics effort could have the greatest measurable impact.

Objective

My objective was to reconstruct the journey at session level, quantify retention and loss between stages, compare outcomes across major segments, and translate the findings into a practical optimisation and experimentation plan.

Analytical Approach

I started with event-level data, where each row records a session interaction such as a home visit, product-page view, cart visit or checkout visit. I reorganised these events into complete session journeys and measured how many sessions reached each successive stage.

How the Data Was Prepared

- **Reconstructed journeys.** I ordered events by timestamp within each session to recover the path from entry to purchase.
- **Validated funnel logic.** I checked stage order, repeated stages, duplicate rows, missing values and attribute consistency within each session.
- **Aggregated at session level.** I calculated the headline funnel and segment rates using distinct sessions rather than raw event rows.
- **Compared key segments.** I compared purchase outcomes by device, region, referral source and six-hour time window.
- **Separated findings from hypotheses.** The funnel shows where friction occurs, while the possible causes still need additional instrumentation or controlled tests.

Dataset and Unit of Analysis

Measure	Value	How it is used
Event records	12,719	Raw interactions used to reconstruct journeys
Sessions	5,000	Primary denominator for funnel and segment rates
Unique users	1,872	Shows that some users generated multiple sessions
Purchase sessions	1,010	Sessions with a purchase outcome
Observation period	January–June 2025	Timestamp range represented in the source data

Data quality checks

Before calculating the funnel, I checked for missing values, duplicate event rows, repeated stages, incorrect stage order and conflicting device, country or referral values within a session. I found no issues in these checks.

Technical Implementation

I used Python to validate the event data, order interactions, rebuild session journeys, calculate funnel metrics and run Pearson chi-square tests. I used Excel to cross-check the results with pivot tables and review the outputs visually. The workflow is repeatable and can be rerun when new event data becomes available.

Funnel Overview

I measured both the number of sessions retained at each stage and the conversion rate from the immediately preceding stage. Looking at both measures made it possible to distinguish a large percentage drop from a large absolute loss of sessions.

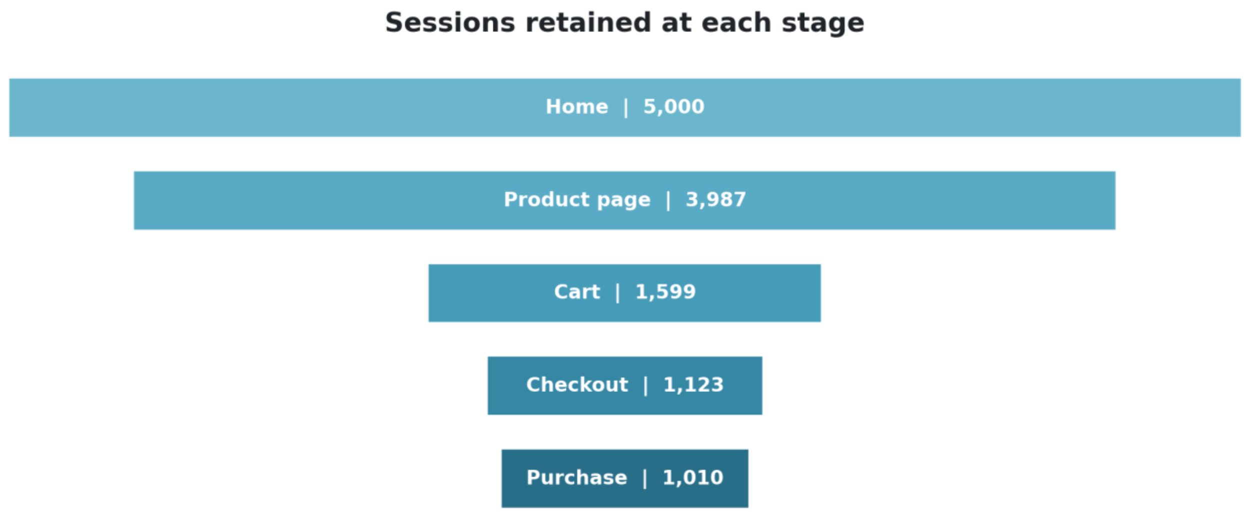

Figure 1. Sessions retained at each stage of the customer journey.

Stage	Sessions	Stage conversion	Drop-off	Sessions lost
Home	5,000	—	—	—
Product page	3,987	79.7%	20.3%	1,013
Cart	1,599	40.1%	59.9%	2,388
Checkout	1,123	70.2%	29.8%	476
Purchase	1,010	89.9%	10.1%	113

What the Funnel Shows

Entry into the product experience is comparatively healthy: 79.7% of home sessions reach a product page. The major loss comes next. Only 40.1% of product-page sessions continue to the cart, removing 2,388 sessions from the journey at a single transition.

Progression improves after cart entry. About 70.2% of cart sessions reach checkout and 89.9% of checkout sessions reach purchase. Later stages still deserve monitoring, but the largest available gain sits earlier, while customers are deciding whether an item is worth adding.

Business interpretation

The issue is more specific than low overall conversion: product page-to-cart progression is weak. This gives product, design and analytics teams one clear metric to improve and monitor.

Product Evaluation Bottleneck

I prioritised the product page-to-cart transition because it has both the largest percentage drop-off and the largest absolute loss. An improvement at this stage would affect more sessions than an equivalent percentage improvement later in the funnel.

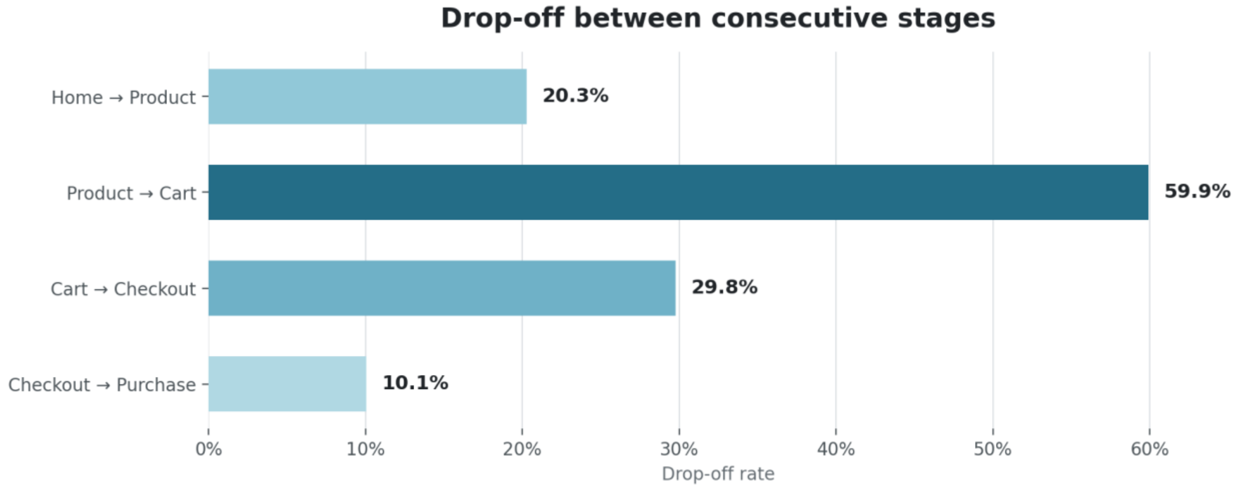

Figure 2. Drop-off rates between consecutive funnel stages.

What I Found

- **The location of friction is clear.** Most loss occurs after a product page is reached but before an item is added to the cart.
- **The loss is large in both relative and absolute terms.** The transition loses 59.9% of eligible sessions, equal to 2,388 sessions.
- **Later-stage completion is comparatively strong.** Once a session reaches checkout, roughly nine out of ten proceed to purchase.
- **The pattern is shared across segments.** Device and regional purchase rates are nearly identical, so I did not isolate one group as the main source of the problem.

Potential Causes to Test

Hypothesis	Controlled test	Primary metric
Pricing clarity	Show total cost, delivery terms and key conditions earlier	Product → cart rate
Value and reassurance	Strengthen benefits, reviews, returns and trust messaging	Product → cart rate
Product information	Test richer imagery, specifications, comparisons and FAQs	Product → cart rate
CTA usability	Test wording, prominence, placement and interaction feedback	Cart entry rate
Variant or stock friction	Instrument selection errors and unavailable combinations	Successful cart additions

How I treated possible causes

The event data shows behaviour, not motivation. I treated pricing, value, trust, information quality and interface design as testable hypotheses rather than proven causes.

Referral Source Analysis

For referral sources, I looked at two separate questions: which source contributed the most purchases, and which source converted its sessions most efficiently. Purchase share reflects both traffic volume and performance, so I did not treat it as a conversion metric on its own.

Referral source	Sessions	Purchases	Purchase rate	Share of purchases
Google	1,280	277	21.64%	27.43%
Email	1,251	251	20.06%	24.85%
Direct	1,226	243	19.82%	24.06%
Social Media	1,243	239	19.23%	23.66%

Interpretation

Google contributed the most purchases at 277 and recorded the highest descriptive purchase rate at 21.64%. Social Media recorded the lowest rate at 19.23%. The difference between the highest and lowest rates was only 2.41 percentage points.

I ran a Pearson chi-square test of independence to check whether referral source and purchase outcome were associated. The result was not statistically significant ($\chi^2(3) = 2.50$, $p = .475$, Cramér's $V = .022$). I therefore treated the ranking as descriptive rather than as evidence for a major budget shift.

Resource Implication

- **Do not increase acquisition spend based on purchase volume alone.** A source may produce more purchases simply because it generated more sessions.
- **Improve the shared product-page experience first.** Every referral source feeds into the same large product-to-cart drop-off.
- **Add cost and value measures before reallocating budget.** Customer acquisition cost, revenue per session and customer value would make the channel comparison more decision-ready.
- **Run channel-specific experiments after improving the baseline.** Landing-page alignment and campaign messaging can then be tested with cleaner measurement.

Decision

I would prioritise conversion efficiency before buying more traffic. Improving a shared bottleneck can benefit every acquisition source, while additional traffic would send more sessions into the same weak transition.

Device and Geographic Patterns

I compared device and regional purchase rates to check whether the overall funnel problem was concentrated in one group. The differences were extremely small, so I focused the recommendation on the common experience rather than a broad device-only or region-only redesign.

Device Comparison

Device	Sessions	Purchases	Purchase rate
Desktop	1,666	339	20.35%
Mobile	1,671	337	20.17%
Tablet	1,663	334	20.08%

Desktop, Mobile and Tablet purchase rates ranged from 20.08% to 20.35%, a spread of only 0.27 percentage points. The Pearson chi-square result was not statistically significant ($\chi^2(2) = 0.04$, $p = .981$, Cramér’s $V = .003$). I therefore did not treat mobile as the primary source of funnel friction.

Represented Regions

Region	Countries represented	Sessions	Purchases	Purchase rate
Europe	France, UK, Germany	2,194	447	20.37%
Americas	USA, Canada	1,421	287	20.20%
Asia-Pacific	India, Australia	1,385	276	19.93%

Europe, the Americas and Asia-Pacific all converted at approximately 20%. The difference was not statistically significant ($\chi^2(2) = 0.10$, $p = .949$, Cramér’s $V = .005$). The data covers France, the UK, Germany, the USA, Canada, India and Australia; because no African countries are represented, I excluded Africa from the regional comparison.

Practical conclusion

The near-identical rates point to a shared experience issue rather than one group performing unusually poorly. I would continue monitoring device and country results but begin with the common product page-to-cart journey.

Time-of-Day Analysis

I grouped session start times into four six-hour periods: late night (00:00–05:59), morning (06:00–11:59), afternoon (12:00–17:59) and evening (18:00–23:59). I then tested whether purchase outcomes differed across these time windows.

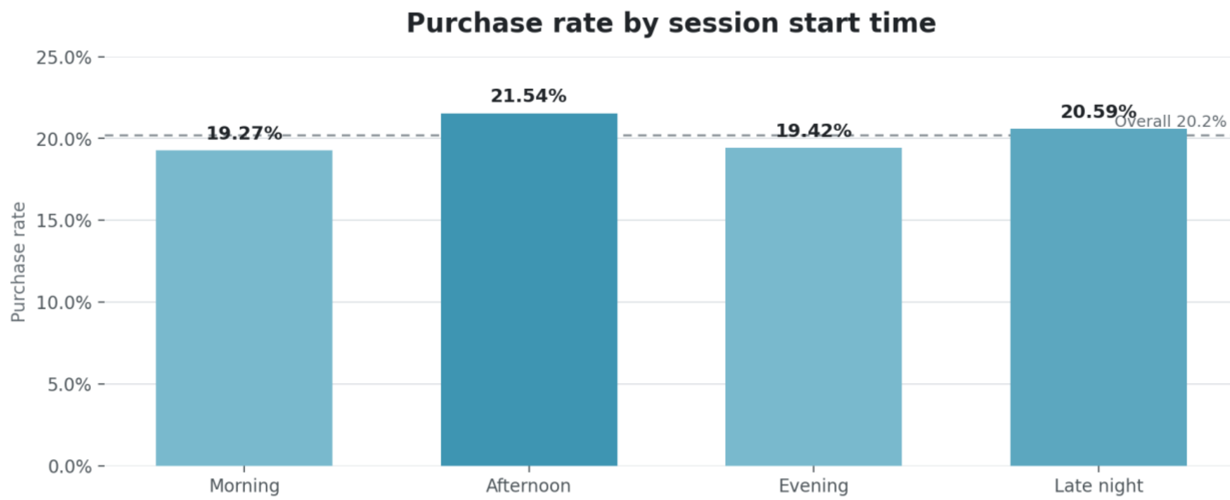

Figure 3. Purchase rate by session start time.

Time segment	Window	Sessions	Purchases	Purchase rate
Morning	06:00–11:59	1,261	243	19.27%
Afternoon	12:00–17:59	1,272	274	21.54%
Evening	18:00–23:59	1,277	248	19.42%
Late night	00:00–05:59	1,190	245	20.59%

Statistical Validation: Pearson Chi-Square

I used a Pearson chi-square test of independence on a 4 × 2 contingency table: four time segments by two outcomes, purchase and no purchase. The null hypothesis was that purchase outcome is independent of session start time. The result was $\chi^2(3) = 2.69$, $p = .442$, with Cramér's V = .023.

The result was not statistically significant because p was greater than .05. Afternoon had the highest descriptive purchase rate at 21.54%, while morning had the lowest at 19.27%, but this difference could reasonably be due to sampling variation. The very small Cramér's V also indicates little practical separation between the groups.

How I used this result

I treated the afternoon result as a hypothesis for an experiment, not as a permanent scheduling rule. The next step would be to test afternoon timing against another window using comparable audiences and a predefined primary metric.

How I Would Test Campaign Timing

- Randomise timing across comparable audiences instead of comparing naturally different traffic periods.
- Measure incremental purchase rate and revenue per session, not only opens or notification clicks.
- Track opt-outs, complaints, repeat engagement and customer value as guardrails.
- Retain the current schedule if the experiment does not produce a reliable and practically meaningful lift.

Recommendations and Resource Prioritisation

I ranked the recommendations by the strength of the evidence and the size of the affected funnel stage. This keeps the action plan focused on measurable opportunities while treating untested explanations as hypotheses rather than established causes.

Priority	Workstream	Evidence strength	Why it comes first
1	Product-page experiments	High	59.9% product-to-cart drop-off and 2,388 lost sessions
2	Cart progression review	Medium	29.8% cart-to-checkout drop-off
3	Measurement and monitoring	High	Current analysis is retrospective
4	Referral and timing tests	Exploratory	Small, non-significant descriptive differences
5	Device or region redesigns	Low	Purchase rates are nearly identical

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Optimise product pages

Make total cost, delivery information, product benefits and reassurance easy to find. Test pricing clarity, product information, trust messaging and CTA presentation separately so the team can learn what actually changes cart entry.

2

Review cart-entry and cart progression

Verify that the add-to-cart interaction is prominent, responsive and accessible. Instrument variant-selection errors, stock problems, unexpected cost exposure and unclear next steps before simplifying the cart journey.

3

Build stage-level monitoring

Track product-to-cart rate as the primary leading metric, alongside absolute session losses, checkout completion, purchase rate, revenue per session and technical performance. Add device, country and referral filters for diagnosis rather than assumption.

4

Validate decisions with controlled tests

Predefine the primary metric, guardrails, sample-size requirement and decision threshold. Use A/B testing or carefully phased experiments rather than relying only on before-and-after comparisons.

Measurement rule

I would only adopt a product-to-cart change if it also protects later-stage completion, revenue quality and customer experience. Both stage movement and final business outcomes should be reported.

Key Outcomes, Limitations and Next Steps

Key Outcomes

- Located the primary funnel bottleneck at the product evaluation stage.
- Quantified the bottleneck as a 59.9% drop-off and 2,388 sessions lost before cart entry.
- Showed that device and represented-region purchase rates are nearly identical.
- Separated descriptive referral and timing differences from statistically reliable effects.
- Produced a repeatable, session-level analysis workflow for future customer-journey data.

Limitations

The data does not include product identity, price, discount exposure, delivery cost, stock status, error logs, experiment assignment, revenue, marketing cost or qualitative feedback. I could therefore identify behavioural friction and prioritise tests, but I could not directly establish why customers left or estimate return on investment.

I used sessions as the primary unit because the funnel describes individual visits. The data contains 1,872 unique users, so a future extension could compare repeat behaviour, first-time versus returning users and longer-term customer value.

What I Would Do Next

- Add product, price, stock, discount, delivery-cost, CTA-click and error-event fields to the event model.
- Create a monitored funnel dashboard with stage rates, absolute losses, segment filters and anomaly alerts.
- Launch the highest-priority product-page experiment and report both stage-level and final purchase outcomes.
- Add revenue, acquisition cost and customer-value measures before changing channel budgets.
- Maintain an experiment log containing the hypothesis, design, decision threshold, result and follow-up action.

Final decision

My first priority is to improve product evaluation and measure product-to-cart movement. Referral, timing, device and regional patterns should remain visible in monitoring, but none of them currently explains the low purchase progression.

Statistical Test Summary

Comparison	χ^2	df	p-value	Cramér's V	Interpretation
Device	0.04	2	.981	.003	No evidence of a device difference
Represented region	0.10	2	.949	.005	No evidence of regional underperformance
Referral source	2.50	3	.475	.022	Directional ranking only
Time of day	2.69	3	.442	.023	Test timing experimentally